INCH-POUND

MIL-PRF-15733/39J
w/AMENDMENT 2
18 November 2005
SUPERSEDING
MIL-PRF-15733/39J
w/AMENDMENT 1
9 December 2003

PERFORMANCE SPECIFICATION SHEET

# FILTERS, RADIO FREQUENCY INTERFERENCE, HERMETICALLY SEALED, STYLE FL91

Part numbers M15733/39-0007, -0008, -0011, -0013, are inactive for new design as of 19 August 1981. See table II for substitution data.

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the filters described herein shall consist of this specification sheet and the latest issue of MIL-PRF-15733.

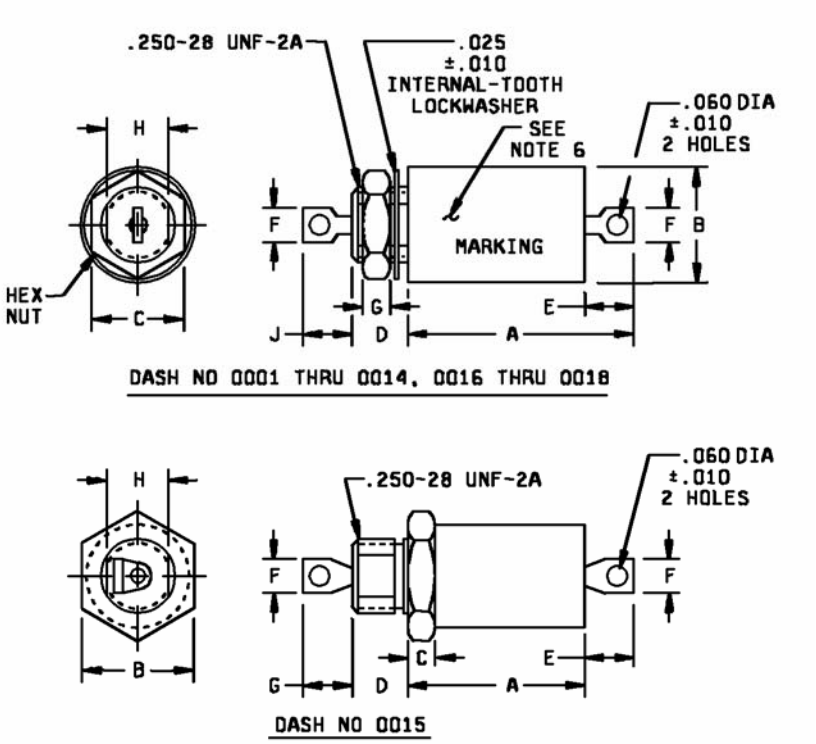



FIGURE 1. Case dimensions and circuit diagrams.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:20Z
Check the source to verify that this is the current version before use.

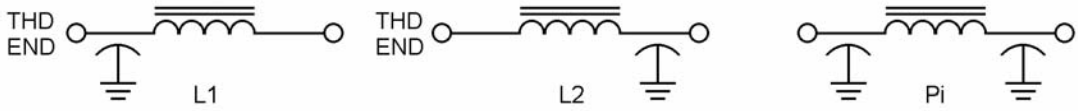
| Dash number | A Max | B Max | C | | D | | E | | F | | G | | H | | J Max | Maximum weight (grams) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | | |
| 0001 | .760 (19.30) | .385 (9.78) | .302 (7.67) | .322 (8.18) | .177 (4.50) | .197 (5.00) | .140 (3.56) | .190 (4.83) | .083 (2.11) | .130 (3.30) | .088 (2.24) | .098 (2.49) | .190 (4.83) | .210 (5.33) | .150 (3.81) | 28.4 |
| 0002 | .760 (19.30) | .385 (9.78) | .302 (7.67) | .322 (8.18) | .177 (4.50) | .197 (5.00) | .140 (3.56) | .190 (4.83) | .083 (2.11) | .130 (3.30) | .088 (2.24) | .098 (2.49) | .190 (4.83) | .210 (5.33) | .150 (3.81) | 28.4 |
| 0003 | .760 (19.30) | .385 (9.78) | .302 (7.67) | .322 (8.18) | .177 (4.50) | .197 (5.00) | .140 (3.56) | .190 (4.83) | .083 (2.11) | .130 (3.30) | .088 (2.24) | .098 (2.49) | .190 (4.83) | .210 (5.33) | .150 (3.81) | 28.4 |
| 0004 | .760 (19.30) | .385 (9.78) | .302 (7.67) | .322 (8.18) | .177 (4.50) | .197 (5.00) | .140 (3.56) | .190 (4.83) | .083 (2.11) | .130 (3.30) | .088 (2.24) | .098 (2.49) | .190 (4.83) | .210 (5.33) | .150 (3.81) | 28.4 |
| 0005 | .760 (19.30) | .385 (9.78) | .302 (7.67) | .322 (8.18) | .177 (4.50) | .197 (5.00) | .140 (3.56) | .190 (4.83) | .083 (2.11) | .130 (3.30) | .088 (2.24) | .098 (2.49) | .190 (4.83) | .210 (5.33) | .150 (3.81) | 28.4 |
| 0006 | .760 (19.30) | .385 (9.78) | .302 (7.67) | .322 (8.18) | .177 (4.50) | .197 (5.00) | .140 (3.56) | .190 (4.83) | .083 (2.11) | .130 (3.30) | .088 (2.24) | .098 (2.49) | .190 (4.83) | .210 (5.33) | .150 (3.81) | 28.4 |
| 0007 1/ | .760 (19.30) | .385 (9.78) | .302 (7.67) | .322 (8.18) | .177 (4.50) | .197 (5.00) | .140 (3.56) | .190 (4.83) | .083 (2.11) | .130 (3.30) | .088 (2.24) | .098 (2.49) | .190 (4.83) | .210 (5.33) | .150 (3.81) | 28.4 |
| 0008 1/ | .760 (19.30) | .385 (9.78) | .302 (7.67) | .322 (8.18) | .177 (4.50) | .197 (5.00) | .140 (3.56) | .190 (4.83) | .083 (2.11) | .130 (3.30) | .088 (2.24) | .098 (2.49) | .190 (4.83) | .210 (5.33) | .150 (3.81) | 28.4 |
| 0009 | .760 (19.30) | .385 (9.78) | .302 (7.67) | .322 (8.18) | .177 (4.50) | .197 (5.00) | .140 (3.56) | .190 (4.83) | .083 (2.11) | .130 (3.30) | .088 (2.24) | .098 (2.49) | .190 (4.83) | .210 (5.33) | .150 (3.81) | 28.4 |
| 0010 | .760 (19.30) | .385 (9.78) | .302 (7.67) | .322 (8.18) | .177 (4.50) | .197 (5.00) | .140 (3.56) | .190 (4.83) | .083 (2.11) | .130 (3.30) | .088 (2.24) | .098 (2.49) | .190 (4.83) | .210 (5.33) | .150 (3.81) | 28.4 |
| 0011 1/ | .760 (19.30) | .385 (9.78) | .302 (7.67) | .322 (8.18) | .177 (4.50) | .197 (5.00) | .140 (3.56) | .190 (4.83) | .083 (2.11) | .130 (3.30) | .088 (2.24) | .098 (2.49) | .190 (4.83) | .210 (5.33) | .150 (3.81) | 28.4 |
| 0012 | .760 (19.30) | .385 (9.78) | .302 (7.67) | .322 (8.18) | .177 (4.50) | .197 (5.00) | .140 (3.56) | .190 (4.83) | .083 (2.11) | .130 (3.30) | .088 (2.24) | .098 (2.49) | .190 (4.83) | .210 (5.33) | .150 (3.81) | 28.4 |
| 0013 1/ | .760 (19.30) | .385 (9.78) | .302 (7.67) | .322 (8.18) | .302 (7.67) | .322 (8.18) | --- | .190 (4.83) | --- | .115 (2.92) | .091 (2.31) | .096 (2.44) | .190 (4.83) | .210 (5.33) | .150 (3.81) | 28.4 |
| 0014 | .760 (19.30) | .385 (9.78) | .302 (7.67) | .322 (8.18) | .302 (7.67) | .322 (8.18) | --- | .190 (4.83) | --- | .115 (2.92) | .091 (2.31) | .096 (2.44) | .190 (4.83) | .210 (5.33) | .150 (3.81) | 28.4 |
| 0015 | .509 (12.93) | .385 (9.78) | .120 (3.05) | .130 (3.30) | .302 (7.67) | .322 (8.18) | --- | .190 (4.83) | --- | .115 (2.92) | --- | 1.50 (3.81) | .190 (4.83) | .210 (5.33) | --- | 28.4 |
| 0016 | .80 (20.3) | .416 (10.57) | --- | --- | .180 (4.57) | .200 (5.08) | --- | .357 (9.07) | .099 (2.51) | .119 (3.02) | --- | --- | .190 (4.83) | .210 (5.33) | --- | 10.0 |
| 0017 | .80 (20.3) | .416 (10.57) | --- | --- | .180 (4.57) | .200 (5.08) | --- | .357 (9.07) | .099 (2.51) | .119 (3.02) | --- | --- | .190 (4.83) | .210 (5.33) | --- | 10.0 |
| 0018 | .80 (20.3) | .416 (10.57) | --- | --- | .180 (4.57) | .200 (5.08) | --- | .357 (9.07) | .099 (2.51) | .119 (3.02) | --- | --- | .190 (4.83) | .210 (5.33) | --- | 10.0 |

1/ Inactive for new design.

FIGURE 1. Case dimensions and circuit diagrams - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:20Z
Check the source to verify that this is the current version before use.




CIRCUIT DIAGRAM

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only.
4. Circuit diagram is for information only.
5. Filters shall be supplied with mounting hardware.
6. Terminal identification (nonsymmetrical filters): The case shall be marked at the threaded end of the filter, with the symbol “C” or the symbol “L”, as follows:

| Circuit | Symbol |
|---|---|
| $L_1$ | C |
| $L_2$ | L |

7. Recommended mounting torque: 48 in-oz.

FIGURE 1. Case dimensions and circuit diagrams - Continued



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:20Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical characteristics.

| Rated voltage (volts dc) | Circuit diagram | Max current (amps) | Max voltage drop (volts) | Capacitance µF GMV | Minimum insertion loss (dB) in accordance with MIL-STD-220 1/ | | | | | | | | | | | | | | | | Dash number |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | AT +25°C | | | | | | | | AT -55°C and +125°C | | | | | | | | |
| | | | | | 30 kHz | 100 kHz | 150 kHz | 300 kHz | 1 MHz | 10 MHz | 100 MHz | 1 GHz | 30 kHz | 100 kHz | 150 kHz | 300 kHz | 1 MHz | 10 MHz | 100 MHz | 1 GHz | |
| 100 | $L_1$ | .5 | 1.0 | .66 | 6 | --- | 28 | 40 | 60 | 60 | 60 | 70 | 4 | --- | 26 | 38 | 58 | 58 | 60 | 70 | 0001 |
| 100 | $L_2$ | .5 | 1.0 | .66 | 6 | --- | 28 | 40 | 60 | 60 | 60 | 70 | 4 | --- | 26 | 38 | 58 | 58 | 60 | 70 | 0002 |
| 100 | $L_1$ | 1.0 | .25 | .45 | 6 | --- | 20 | 30 | 50 | 60 | 60 | 70 | 4 | --- | 18 | 28 | 48 | 58 | 60 | 70 | 0003 |
| 100 | $L_2$ | 1.0 | .25 | .45 | 6 | --- | 20 | 30 | 50 | 60 | 60 | 70 | 4 | --- | 18 | 28 | 48 | 58 | 60 | 70 | 0004 |
| 100 | $L_1$ | 2.0 | .14 | .45 | 6 | --- | 20 | 26 | 40 | 60 | 60 | 70 | 4 | --- | 18 | 24 | 38 | 58 | 60 | 70 | 0005 |
| 100 | $L_2$ | 2.0 | .14 | .45 | 6 | --- | 20 | 26 | 40 | 60 | 60 | 70 | 4 | --- | 18 | 24 | 38 | 58 | 60 | 70 | 0006 |
| 100 | $L_1$ | 5.0 | .075 | .45 | 6 | --- | 20 | 26 | 35 | 60 | 60 | 70 | 4 | --- | 18 | 24 | 33 | 58 | 60 | 70 | 0007 2/ |
| 100 | $L_2$ | 5.0 | .075 | .45 | 6 | --- | 20 | 26 | 35 | 60 | 60 | 70 | 4 | --- | 18 | 24 | 33 | 58 | 60 | 70 | 0008 2/ |
| 100 | $L_1$ | 10.0 | .03 | .45 | 6 | --- | 20 | 26 | 35 | 56 | 60 | 70 | 4 | --- | 18 | 24 | 33 | 54 | 60 | 70 | 0009 |
| 100 | $L_2$ | 10.0 | .03 | .45 | 6 | --- | 20 | 26 | 35 | 56 | 60 | 70 | 4 | --- | 18 | 24 | 33 | 54 | 60 | 70 | 0010 |
| 100 | $P_i$ | 10.0 | .03 | .99 | 9 | --- | 24 | 29 | 40 | 70 | 70 | 70 | 7 | --- | 22 | 27 | 38 | 68 | 70 | 70 | 0011 2/ |
| 100 | $P_i$ | 10.0 | .03 | .99 | 9 | --- | 24 | 29 | 40 | 70 | 70 | 70 | 7 | --- | 22 | 27 | 38 | 68 | 70 | 70 | 0012 |
| 100 | $P_i$ | 10.0 | .03 | 1.5 | 9 | --- | 24 | 29 | 40 | 70 | 70 | 70 | 7 | --- | 22 | 27 | 38 | 68 | 70 | 70 | 0015 |
| 100 | $P_i$ | .25 | .38 | 1.5 | --- | 58 | --- | 76 | 80 | 80 | 80 | 80 | --- | 56 | --- | 74 | 78 | 78 | 80 | 80 | 0016 |
| 100 | $P_i$ | 3.0 | .18 | 1.5 | --- | 14 | --- | 41 | 76 | 80 | 80 | 80 | --- | 12 | --- | 39 | 74 | 78 | 80 | 80 | 0017 |
| 100 | $P_i$ | 5.0 | .20 | 1.5 | --- | --- | --- | 28 | 68 | 80 | 80 | 80 | --- | --- | --- | 26 | 66 | 78 | 80 | 80 | 0018 |
| 50 | $L_2$ | 10.0 | .03 | 1.4 | 14 | --- | 28 | 34 | 44 | 52 | 70 | 70 | 12 | --- | 26 | 32 | 42 | 50 | 70 | 70 | 0013 2/ |
| 50 | $L_1$ | 10.0 | .03 | 1.4 | 14 | --- | 28 | 34 | 44 | 52 | 70 | 70 | 12 | --- | 26 | 32 | 42 | 50 | 70 | 70 | 0014 |

1/ Insertion loss measurements shall be made under full load over the frequency range of 100 kHz to 10 MHz. Insertion loss measurements above or below this frequency range shall be made under no load.

2/ Inactive for new design.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:20Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Weight: See figure 1.

Case: Metal.

Case and mounting hardware finish: In accordance with MIL-PRF-15733. Pure tin finish is prohibited.

Seal: Metal-to-glass.

Terminals: Solderable.

Operating temperature range: -55°C to +125°C.

Rated voltage: See table I.

Rated current: See table I.

Insertion loss: In accordance with MIL-PRF-15733 and table I.

Seal: In accordance with MIL-PRF-15733.

Capacitance to ground: In accordance with MIL-PRF-15733 and table I.

Dissipation factor: 3.0 percent maximum.

Temperature rise: 25°C, maximum.

Barometric pressure (reduced): In accordance with MIL-PRF-15733 and method 105, MIL-STD-202, test condition D (100,000 ft) for -0001 through -0010; Test condition C (70,000 ft) for -0011 through -0015; and test condition B (50,000 ft) for -0016 through -0018.

Insulation resistance: In accordance with MIL-PRF-15733 and the following:

At 25°C the insulation resistance shall not be less than 1000 megohms.

At +125°C the insulation resistance shall not be less than 100 megohms.

Voltage drop: See table I.

Terminal strength: In accordance with MIL-PRF15733 and method 211, MIL-STD-202; test condition A (pull).

Force: 5 pounds.

Salt atmosphere (corrosion): In accordance with MIL-PRF-15733 and method 101, MIL-STD-202; test condition A.

Shock (specified pulse): In accordance with MIL-PRF-15733 and method 213, MIL-STD-202. The following details apply:

Test condition I - (-0001 thru -0014 and -0016 through -0018).

Test condition A - (-0015).



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:20Z
Check the source to verify that this is the current version before use.

Vibration, high frequency: In accordance with MIL-PRF-15733 and method 204, MIL-STD-202; test condition D (20G). The following detail shall apply:

Temperature conditioning: Prior to vibration, filters shall be heated in an oven to +125°C plus one-half the temperature rise. Vibration in each plane shall begin within 5 minutes after removal of filters from the oven.

* Life: In accordance with MIL-PRF-15733 and method 108, MIL-STD-202; test condition D (1,000 hrs) for qualification; test condition B (250 hrs) for group C inspection.

Part or Identifying Number (PIN): M15733/39-(dash number from table I).

Extension of qualification:

Qualification testing and approval to M15733/39-0014 shall be sufficient to grant qualification approval to M15733/39-0013 and M15733/39-0014.

Qualification testing and approval to M15733/39-0012 and M15733/39-0016 shall be sufficient to grant qualification approval to M15733/39-0001 through -0012 and M15733/39-0015 through -0018.

Substitution data: See table II.

Table II. Substitution data.

| Inactive PIN | Superseding specification | Superseding PIN |
|---|---|---|
| M15733/39-0007 | MIL-PRF-28861/2 | M28861/02-010TB |
| M15733/39-0008 | MIL-PRF-28861/2 | M28861/02-011TB |
| M15733/39-0011 | MIL-PRF-15733/39 | M15733/39-0012 |
| M15733/39-0013 | MIL-PRF-28861/1 | M28861/01-001TB |

NOTES:

* Referenced documents. In addition to MIL-PRF-15733, this specification sheet references the following documents:

MIL-PRF-28861/1
MIL-PRF-28861/2
MIL-STD-202
MIL-STD-220

Amendment notations. The margins of this specification sheet are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:20Z
Check the source to verify that this is the current version before use.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5915-2005-005)

Review activities:
Army - AT, AV
Navy - AS, CG, MC, OS, SH
Air Force – 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:20Z
Check the source to verify that this is the current version before use.